(ii)

$$\text{Total energy required} = \left(E_{\text{climb}} + E_{\text{speed up}}\right) \times \frac{100}{24} \quad \text{[C1]}$$

$$= \left[\left(5.67 \times 10^{8}\right) + \left(9.74 \times 10^{7}\right)\right] \times \frac{100}{24}$$

$$= 2.768 \times 10^{9}\ \text{J} \quad \text{[A1:correct calculation]}$$

$$\text{mass of fuel} = \frac{\text{total energy}}{\text{fuel energy density}} \quad \text{[C1]}$$

$$= \frac{2.768 \times 10^{9}}{45 \times 10^{6}}$$

$$= 61.5\ \text{kg} \quad \text{[A1: correct calculation]}$$

(iii) The mass of fuel burnt/lost is very small compared to the operating mass of the aircraft. [B1]

(d) **(i)**

u / m s^{-1}	h / m	ρ / kg m^{-3}	$u^2\rho$ / kg m^{-1} s^{-2}	D / ×10^6 N
160	5500	0.698	17900	1.160
167	5730	0.681	19000	1.221